The problem: How to store an n-d pdf?

- We are making excellent progress in the development of background to the particle filter!
  - All of the basic math behind approximating integrals is done!
- One remaining concept relates to how we will represent and store the pdfs that are recursively updated: $f(x_k \mid \mathbb{Z}_{k-1})$ and $f(x_k \mid \mathbb{Z}_k)$.
- In principle, storing a 1-d pdf $f(x)$ as a function of x (e.g., as a table) requires an infinite number of entries (values of x and $f(x)$).
- Storing a multidimensional pdf requires storing $f(x)$ for every possible x, a multiply infinite table.
- This clearly isn't possible, and is not a feasible approach even if we very coarsely quantize each dimension of x to a few samples.
- We need a better approach.

Multi-dimensional Dirac delta function

- We'll use the multidimensional Dirac delta (impulse) function.
- A 1-d Dirac delta is a "generalized" function defined by:

$$\int_{-\infty}^{\infty} \delta(x)\,\mathrm{d}x = 1 \quad \text{and} \quad \delta(x-a) = 0\;\forall\, x \neq a.$$

- From these properties, we can show that: $f(a) = \int_{-\infty}^{\infty} f(x)\delta(x-a)\,\mathrm{d}x$.
  - This is called the "sifting property" of the Dirac delta function.
- A multivariable impulse is defined as

$$\delta(x) = \prod_i \delta(x_i) = \delta(x_0)\delta(x_1)\cdots\delta(x_n),$$

  where x_i is the ith element of vector x, which has n elements in total.
- Particle filters represent $f(x_k \mid \mathbb{Z}_k)$ *mathematically* as a sum of n-d impulses.
  - $f(x_k \mid \mathbb{Z}_k)$ is used *computationally* in a different format, as you'll learn next week.

Integrating with the Dirac delta function

- To investigate, consider a simple example with pdf $f(x)$.
- We model the pdf as $\hat{f}(x) = \frac{1}{N}\sum_{i=1}^{N}\delta(x - x^{(i)})$, where $x^{(i)}$ are IID samples drawn from $f(x)$; we replace $f(x)$ by $\hat{f}(x)$ when finding a mean:

$$\mu = \int_{-\infty}^{\infty} g(x) f(x)\,\mathrm{d}x$$

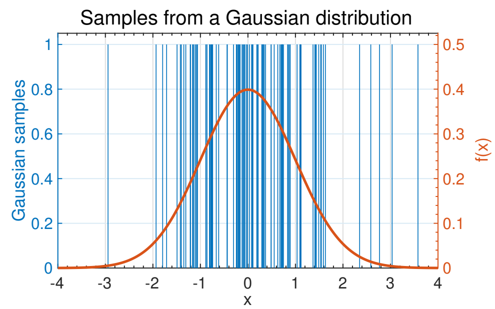
$$\hat{\mu} = \int_{-\infty}^{\infty} g(x)\hat{f}(x)\,\mathrm{d}x = \int_{-\infty}^{\infty} g(x)\frac{1}{N}\sum_{i=1}^{N}\delta(x - x^{(i)})\,\mathrm{d}x$$

$$= \frac{1}{N}\sum_{i=1}^{N}\int_{-\infty}^{\infty} g(x)\delta(x - x^{(i)})\,\mathrm{d}x = \frac{1}{N}\sum_{i=1}^{N} g(x^{(i)}).$$

- So, using our approximation $\hat{f}(x)$, we arrived at the same estimator as derived earlier based on Monte-Carlo integration.

Are $f(x)$ and $\hat{f}(x)$ the same thing?

- But, is it fair to say $\hat{f}(x) \approx f(x)$? Suppose $f(x)$ is Gaussian.
- The figure draws $N = 100$ impulses sampled from $f(x)$ without arrows (for clarity), superimposed on $f(x)$.
- It is impossible to argue any equivalence except under an integral.
- This is a kind of scatter plot, but if you were given only the impulses it would have been tricky to guess the true shape of the distribution. (But, see next lesson.)
- In some sense, $f(x)$ and $\hat{f}(x)$ are nothing alike.



What is the same about $f(x)$ and $\hat{f}(x)$?

- While $f(x)$ and $\hat{f}(x)$ are completely different in some sense, they *are alike* in that:

$$\mu = \int_{-\infty}^{\infty} g(x) f(x)\,\mathrm{d}x \approx \int_{-\infty}^{\infty} g(x)\hat{f}(x)\,\mathrm{d}x.$$

  - So, we can calculate mean using $\hat{f}(x)$ instead of $f(x)$, if we like.
- It is also true that the cumulative distribution function (cdf) matches:

$$F(x) = \int_{-\infty}^{x} f(\alpha)\,\mathrm{d}\alpha \approx \int_{-\infty}^{x} \hat{f}(\alpha)\,\mathrm{d}\alpha.$$

  - So, we can calculate percentiles (including median) from $\hat{f}(x)$.
- But, there is no obvious way to calculate mode of $f(x)$ from $\hat{f}(x)$.[1]
- For most applications, representing pdfs using Dirac deltas (impulses) is fine.

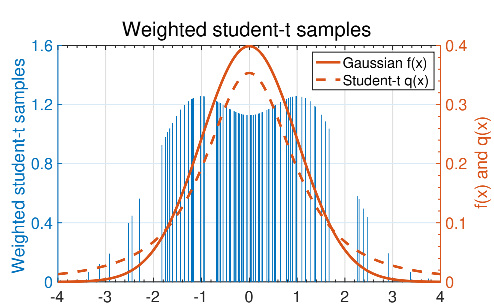
[1] Can use Viterbi particle filters — see Prof. McNames' course notes and videos for details.

Modification to consider importance density

- Final issue: We won't be able to draw samples from $f(x)$, but will instead draw samples from importance density $q(x)$.
- The equivalent effect on the estimated pdf is to weight it

$$\hat{f}(x) = \sum_i w^{(i)}\delta(x - x^{(i)}), \quad \text{where} \quad w^{(i)} = f(x^{(i)})/q(x^{(i)}).$$

- This converges in the same manner as the unweighted approximation of the pdf.
- In the example, $q(x)$ is the student-t distribution with two degrees of freedom.
- In general, $f(x^{(i)}) \neq q(x^{(i)})$, so $w^{(i)} \neq 1$.
- This $\hat{f}(x)$ still approximates the true $f(x)$ well in a Monte–Carlo sense, especially when the number of samples N is large.



Summary

- It is not feasible to store the pdfs required by a particle filter as multidimensional lookup tables (LUTs).
- We instead represent pdfs as sums of multidimensional Dirac delta impulse functions.
- The locations of these impulses are determined from values sampled from the pdfs.
  - Points are densest in high-probability regions and sparsest in low-probability regions.
- When the number of samples N is large, the math using Dirac delta functions converges to the math we have already seen regarding Monte–Carlo integration.
- From now on, we will represent pdfs *mathematically* as summations of impulses (weighted summations if we are using an importance density).
- We will use these pdfs *computationally* by storing the locations (and weights) of the impulses—a much smaller table than if we were to make an LUT for the entire pdf.